Forces
Newton's Laws of Motion

Reading Assignment:

Chapter 4

Summary

Until now: *Kinematics* = description of motion with constant acceleration without reference to what causes acceleration.

Next: **Newtonian Mechanics**

Acceleration is due to the action of *Forces*.

- Applicable to motion of galaxies, stars, planets, aircraft, footballs, dust particles moving at speeds small compared to light.

Forces

- What is a *force*?
- e.g., push a car,
- pull a sleigh with a rope,
- friction, air resistance.
- = “action at a distance”
- **<u>A Force acting on a body causes the body to *accelerate.*</u>**
- Forces cause *acceleration.*
- *The unit of Force is the Newton (kgm/s^2)*

Forces

- All forces cause a change in motion of an object.
- It is a vector quantity so if a number of forces act on a body together….
  - *The net force is the algebraic sum of all the forces.*
- *Sir Isaac Newton devised 3 laws to describe forces*

Fundamental Forces

- Just 3 forces account for all known influences

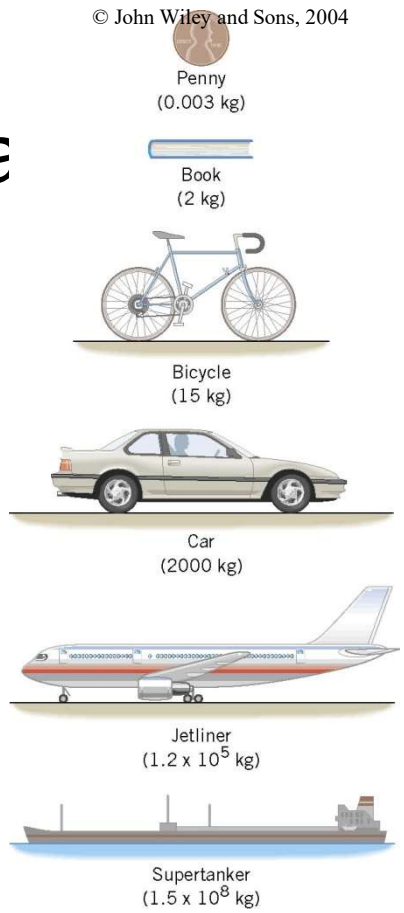
1. Gravitational force
2. Electro weak force
   - Electrical & magnetic forces
   - Weak nuclear force (radioactivity)
3. Strong nuclear force

Newton's First Law of Motion

- "An object continues in a state of rest or in a state of motion at a constant speed along a straight line, unless compelled to change that state by the action of a net force."

Inertia and mass

- *Inertia* is the natural tendency of a body to remain at rest or in uniform motion with constant speed along a straight line.
- SI unit of mass (and inertia): kilogram (kg)



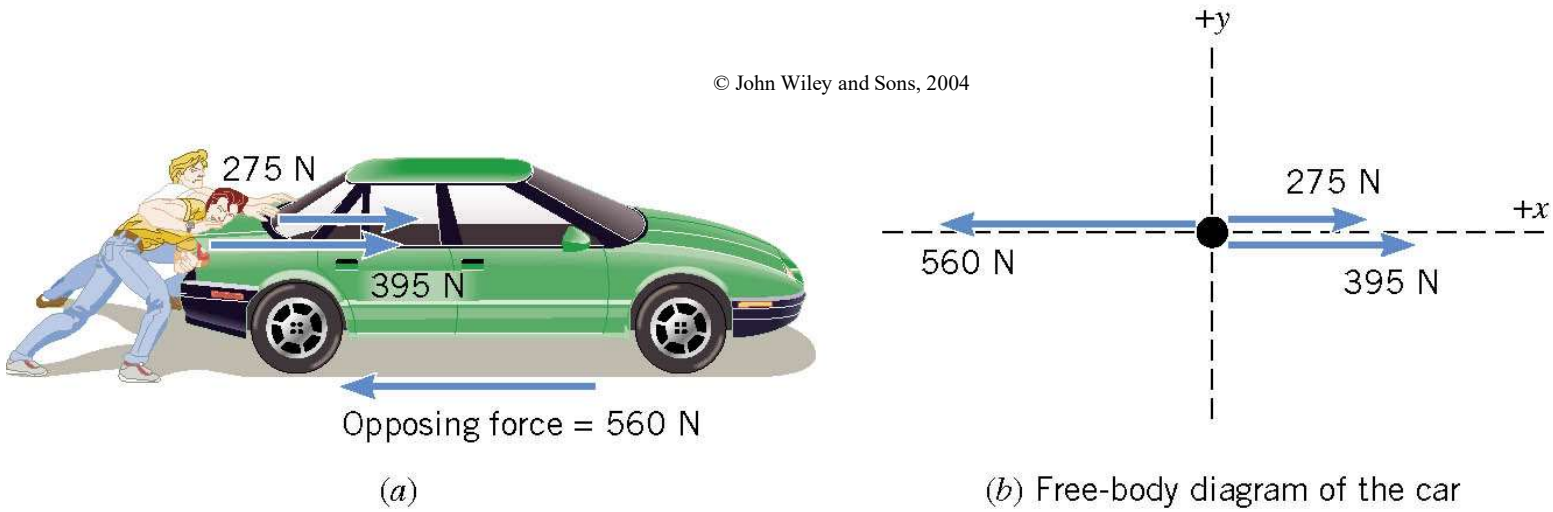
The mass of an object is a quantitative measure of its inertia.

Newton's Second Law of Motion

- "When a net external force $\Sigma \underline{F}$ acts on an object of mass ***m***, the acceleration $\underline{\boldsymbol{a}}$ that results is directly proportional to the net force and has a magnitude that is inversely proportional to the mass. The direction of the acceleration is the same as that of the net force."
- SI unit of Force: Newton (N)= 1 kg.m/s^2

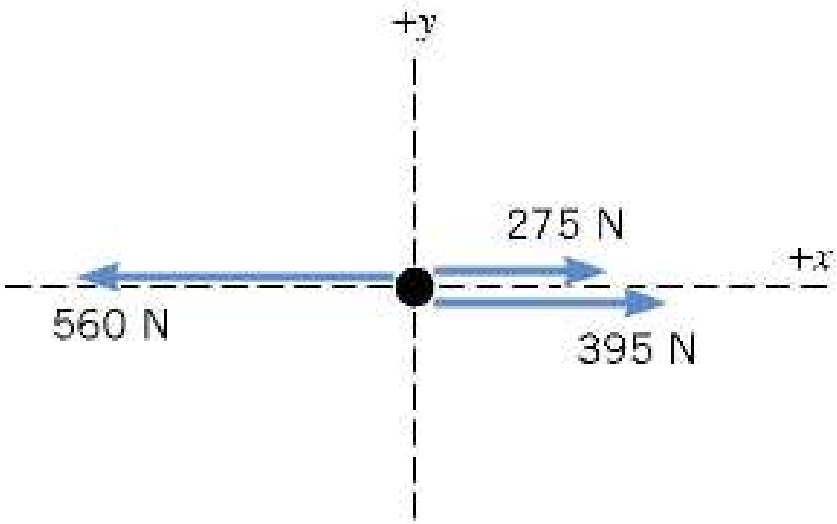
$$\underline{a} = \frac{\Sigma \underline{F}}{m} \quad \text{or} \quad \Sigma \underline{F} = m\,\underline{a}$$

Free body diagrams



- Car has mass 1850kg. One person applies force of 275 N, another applies force of 395 N. Frictional force of 560 N acts in opposite direction. What is acceleration of car?

(a)



(b) Free-body diagram of the car

The net force in this case is:

275 N + 395 N – 560 N = +110 N

and is directed along the + x axis of the coordinate system.

If the mass of the car is 1850 kg then, by Newton's second law, the acceleration is

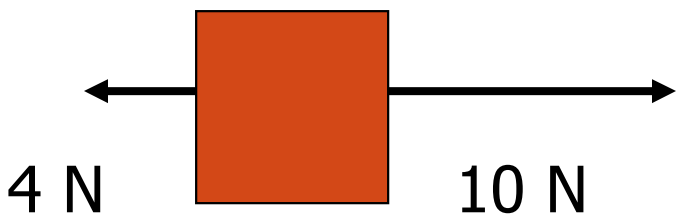
$$a = \frac{\sum F}{m} = \frac{+110\,\mathrm{N}}{1850\,\mathrm{kg}} = +0.059\,\mathrm{m/s^2}$$

The net force on an object is the vector sum of all forces acting on that object.

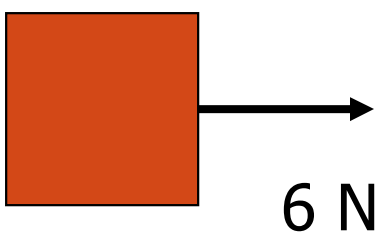
The SI unit of force is the Newton (N).

$$(\mathrm{kg})\left(\frac{\mathrm{m}}{\mathrm{s}^2}\right)=\frac{\mathrm{kg}\cdot\mathrm{m}}{\mathrm{s}^2}$$

<u>Individual Forces</u>

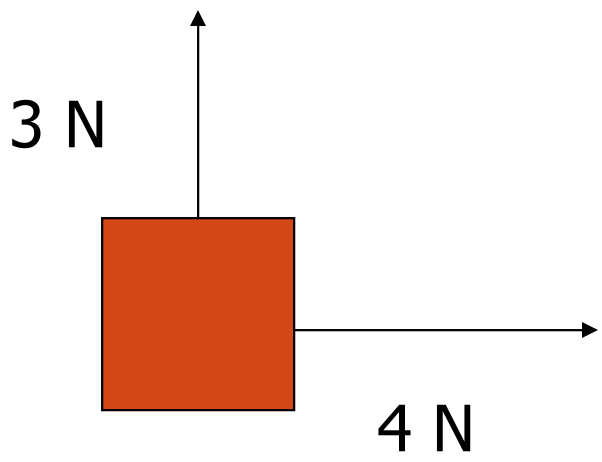


<u>Net Force</u>

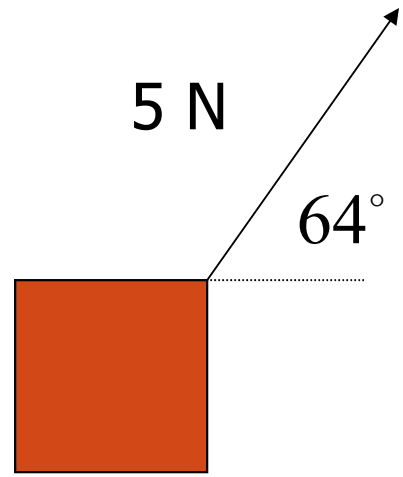


- Consider any two directions at right angles, e.g. horizontal (x) and vertical (y).
- A force acting in the x direction causes acceleration in the x direction only.
  - i.e., $F_x = ma_x$
- A force acting in the y direction causes acceleration in the y direction only.
  - i.e., $F_y = ma_y$

Individual Forces

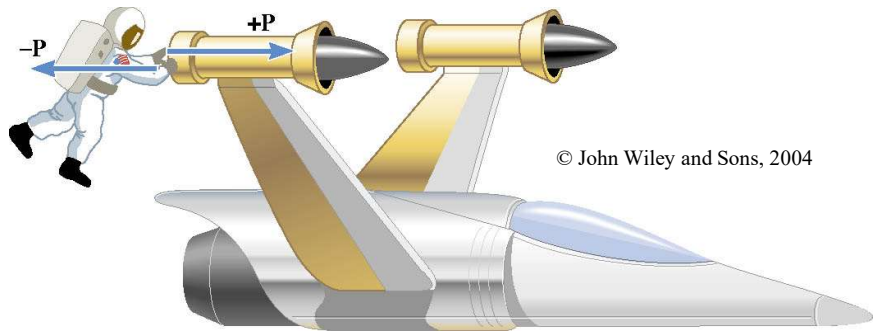


Net Force



Newton's Third Law of Motion

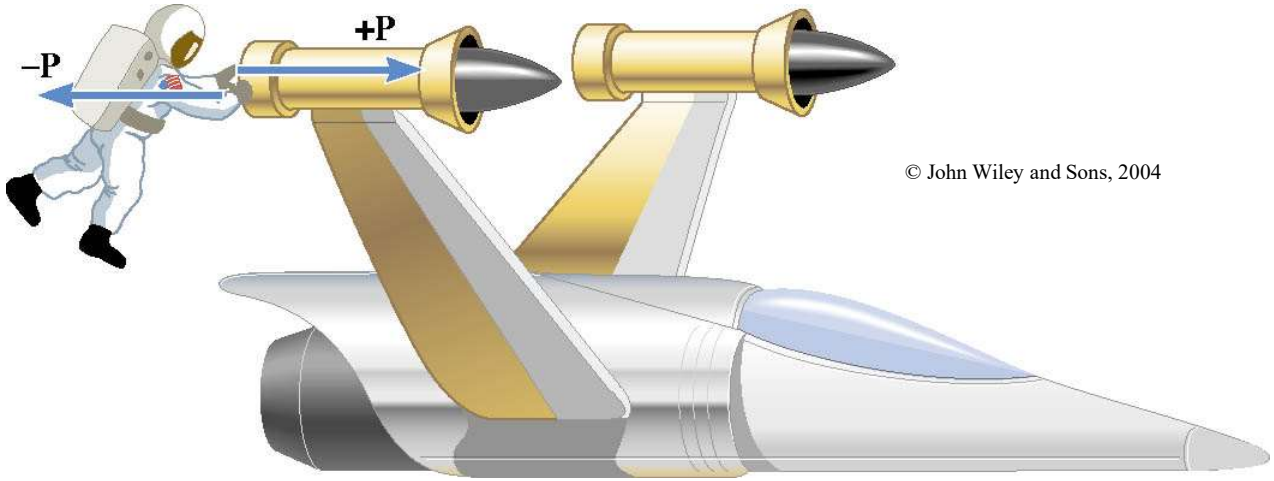
- "Whenever one body exerts a force on a second body, the second body exerts an oppositely directed force of equal magnitude on the first body."
- **or**
- action - reaction law.



"For every action (force), there is an equal and opposite reaction."

Newton's Third Law of Motion

- Mass of spacecraft = 11000 kg. Mass of astronaut = 92 kg. Astronaut pushes with force +36 N. What accelerations result?



"For every action (force), there is an equal and opposite reaction."

On the spacecraft $\sum \vec{\mathbf{F}} = \vec{\mathbf{P}}$.

On the astronaut $\sum \vec{\mathbf{F}} = -\vec{\mathbf{P}}$.

$$\vec{\mathbf{a}}_s = \frac{\vec{\mathbf{P}}}{m_s} = \frac{+36\ \mathrm{N}}{11{,}000\ \mathrm{kg}} = +0.0033\ \mathrm{m/s^2}$$

$$\vec{\mathbf{a}}_A = \frac{-\vec{\mathbf{P}}}{m_A} = \frac{-36\ \mathrm{N}}{92\ \mathrm{kg}} = -0.39\ \mathrm{m/s^2}$$

Summary

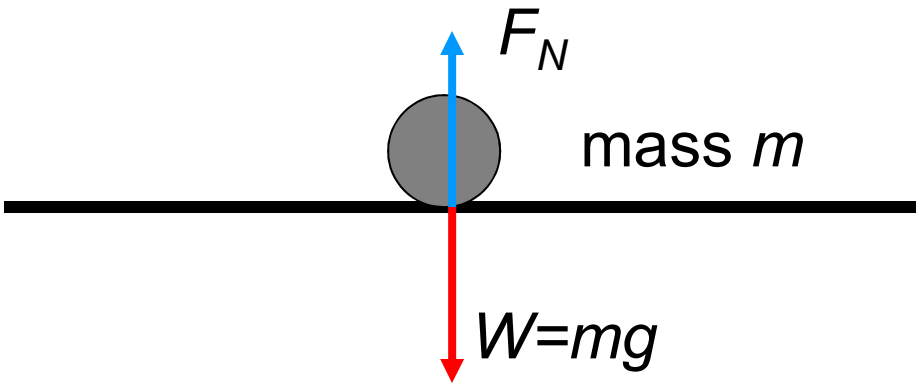
- All changes in motion are caused by the action of *Forces*.
- Specifically,Forces cause *acceleration.*
- Mass of a body is quantitative measure of how difficult it is to cause it to accelerate.

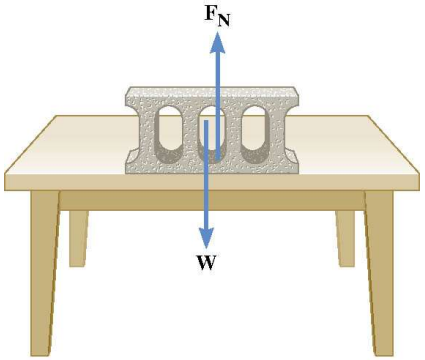
Equilibrium and the Normal Force

- A man standing on the surface of the earth is clearly not accelerating downwards at 9.8 m/s^2.

Gravitational force still acts, but a = 0 m/s^2.

There must be another force balancing the gravitational force - “The Normal Force”. The net force acting on the body is zero, and we say it is “in equilibrium”





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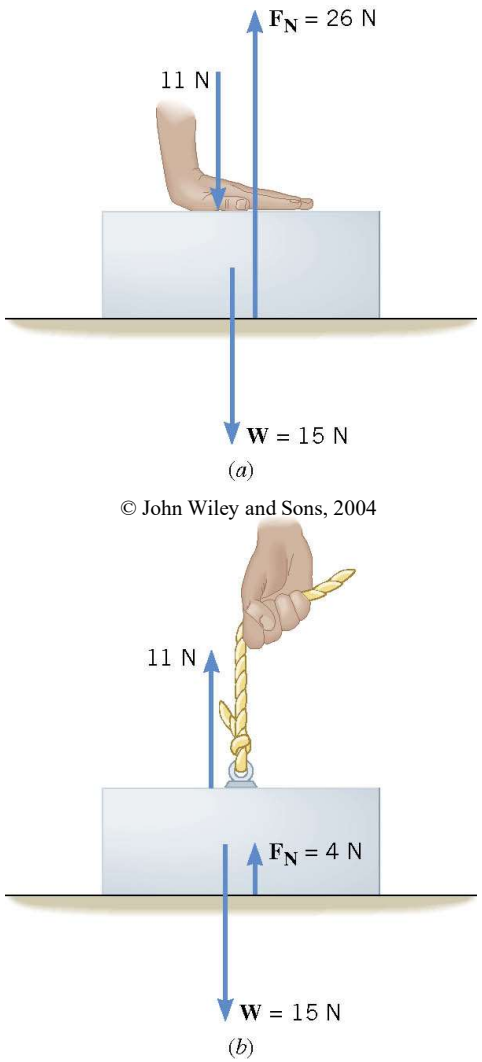
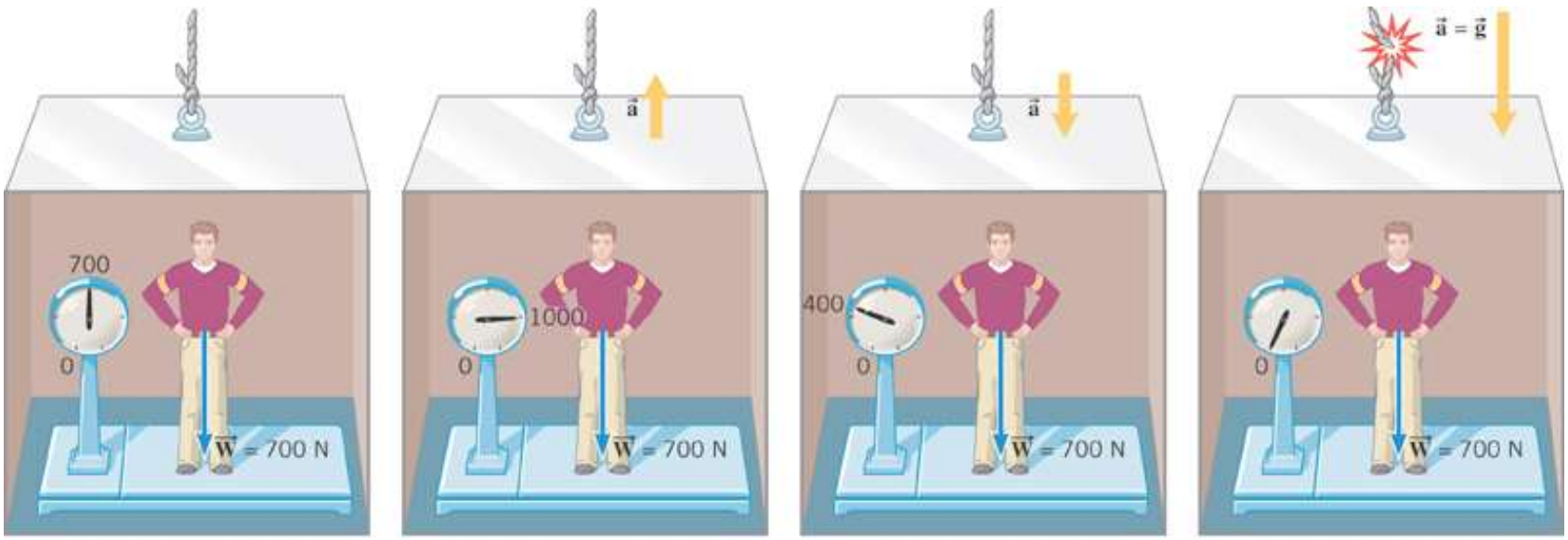


- The normal force $\boldsymbol{F_N}$ is one component of the force that a surface exerts on an object with which it is in contact – the component that is perpendicular to the surface.
- Newton's 3rd law –to very force…..

Apparent Weight

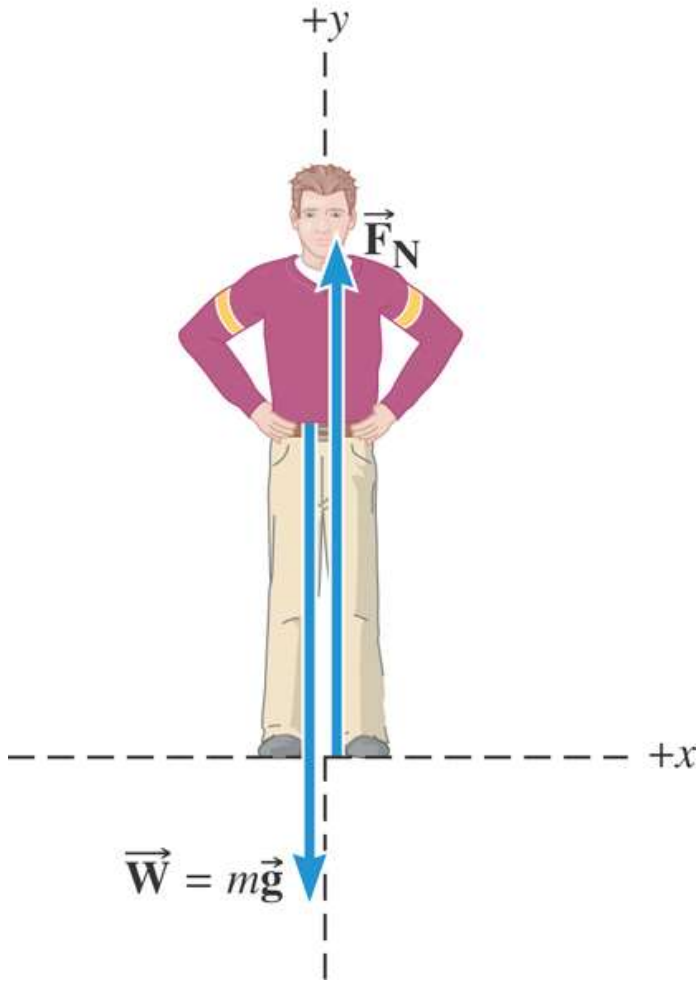
The apparent weight of an object is the reading of the scale.

It is equal to the normal force the man exerts on the scale.



(*a*) No acceleration ($\vec{v}$ = constant) (*b*) Upward acceleration (*c*) Downward acceleration (*d*) Free-fall

4.8 The Normal Force



$$\sum F_y = +F_N - mg = ma$$

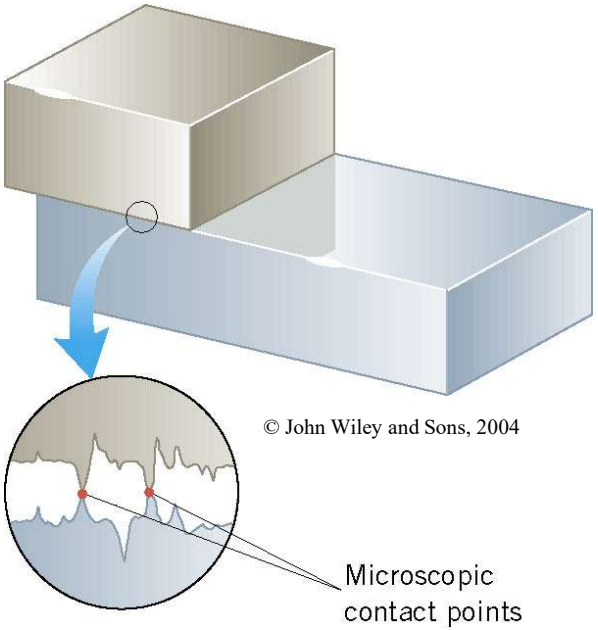
$$F_N = mg + ma$$

apparent weight

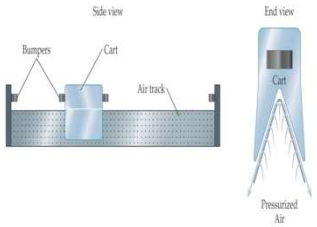
true weight

Frictional Force

- Frictional Force acts to oppose motion along a surface. Due to the roughness of the touching surfaces
- Smoother surface means less friction.
- We need to apply a force to a stationary body to overcome static friction.
- Once moving friction always opposes motion!

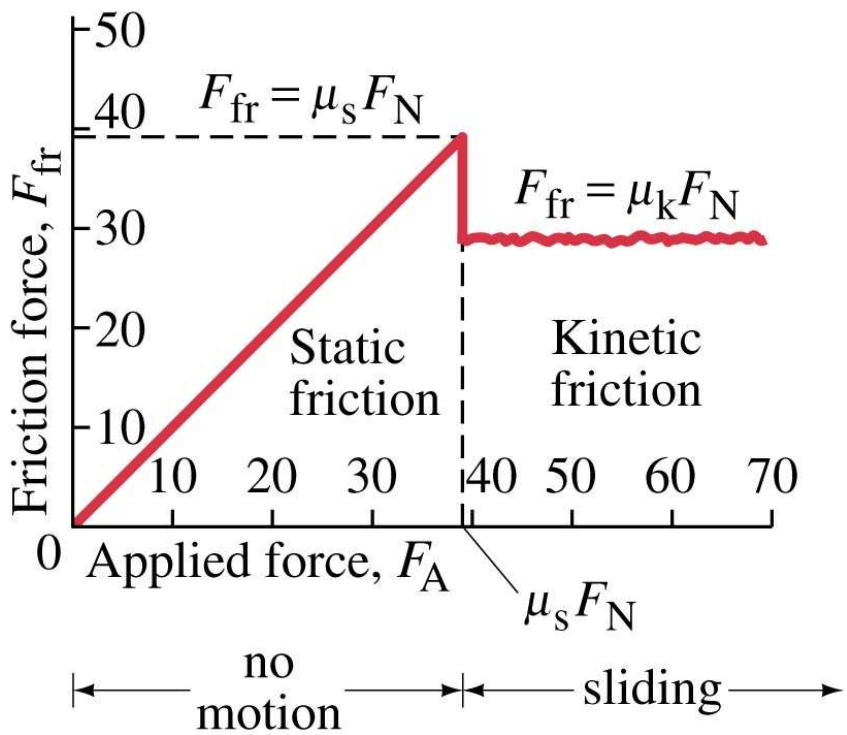


Friction in the real world



4-8 Problems Involving Friction,

The static frictional force increases as the applied force increases, until it reaches its maximum. Then the object starts to move, and the kinetic frictional force takes over.



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Frictional Force

- <u>Empirical model</u>

Static case: Static friction force < $\mu_s F_N$

Kinetic case (Moving body):

Kinetic frictional force = $\mu_k F_N$

F_N is the normal force

μ_s = coefficient of static friction

μ_k = coefficient of kinetic friction

Generally,

$$\mu_s > \mu_k$$

Frictional Force

- To just get a stationary body moving we need to apply a force equal to than $f_s^{MAX} = \mu_s F_N$

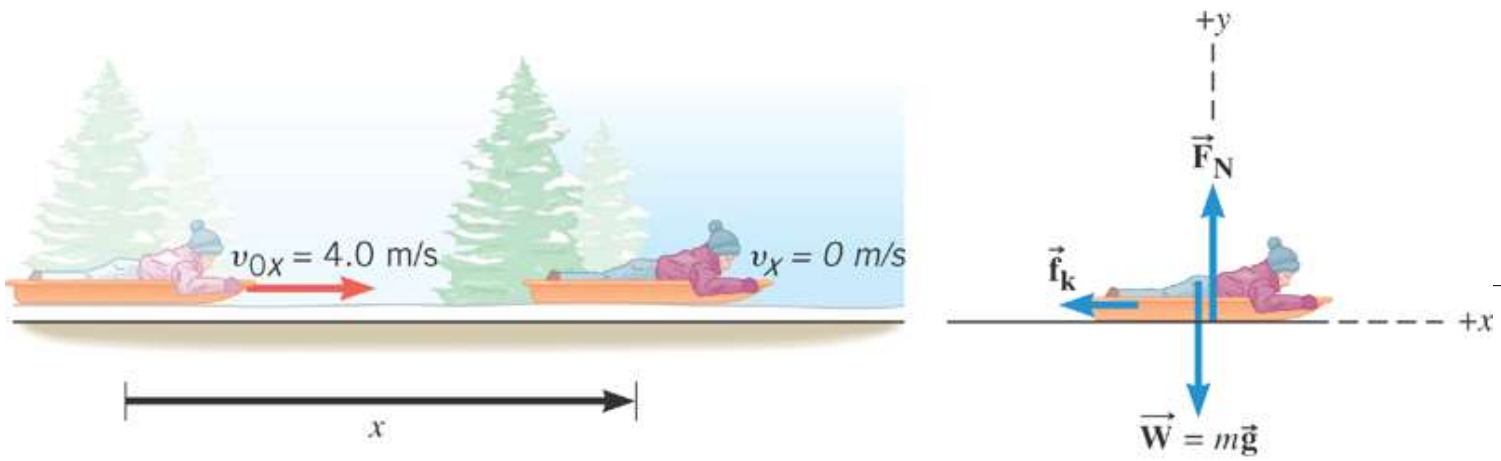
$$\mu_s = \frac{f_s^{MAX}}{F_N}$$

- Once moving we have kinetic frictional forces equal to $f_k = \mu_k F_N$
- The size of μ_k determines how long the object will move
- Compare sliding on ice and dry concrete!

Table 6-1
Typical Coefficients of Friction

Materials	Kinetic, μ_k	Static, μ_s
Rubber on concrete (dry)	0.80	0.90
Steel on steel	0.57	0.74
Glass on glass	0.40	0.94
Wood on leather	0.40	0.50
Copper on steel	0.36	0.53
Rubber on concrete (wet)	0.25	0.30
Steel on ice	0.06	0.10
Waxed ski on snow	0.05	0.10
Teflon on Teflon	0.04	0.04

4.9 Static and Kinetic Frictional Forces



(a) (b) Free-body diagram for the sled and rider

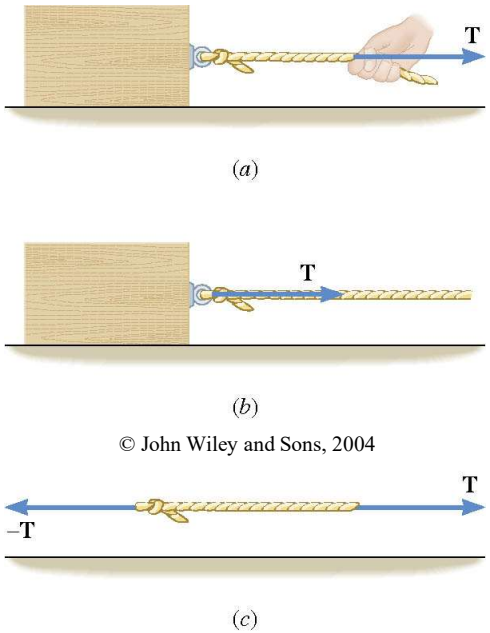
Suppose the coefficient of kinetic friction is 0.05 and the total mass is 40kg. What is the kinetic frictional force?

$$f_k = \mu_k F_N = \mu_k mg =$$

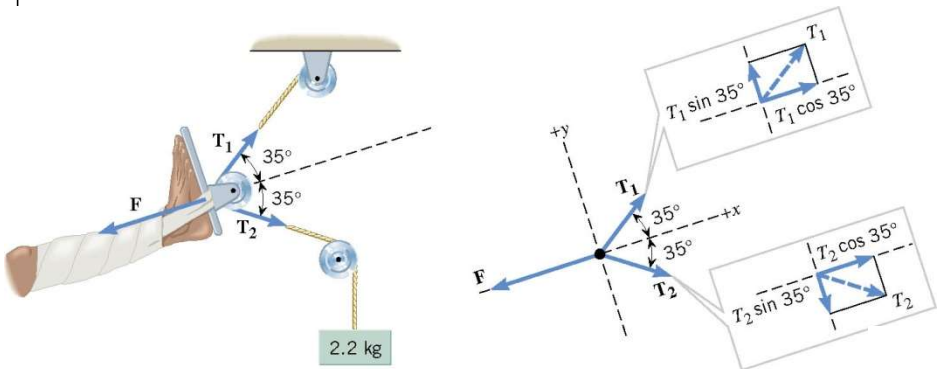
$$0.05(40\text{kg})(9.80\,\text{m/s}^2) = 20\text{N}$$

The Tension Force

- An object is in equilibrium when it has zero acceleration.
- This is important when a number of forces act on a body together.
- Below the force T is applied to the box because of the tension in the rope.
- Use the concept of massless rope - examples



The Tension Force



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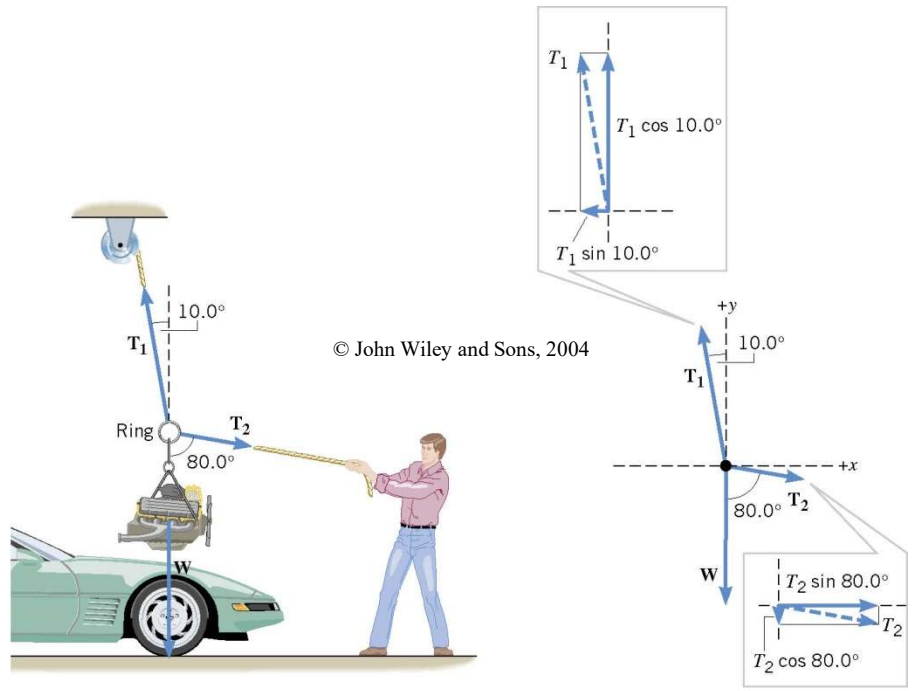
(a) (b) Free-body diagram for the foot pu



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(a) (b) Free-body diagram for the ring

Forces Newton's Law of Universal Gravitation

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So far...

- All changes in motion are caused by the action of *Forces*.

Forces cause *acceleration:*

$$\underline{a} = \frac{\Sigma \underline{F}}{m}$$

3 fundamental forces

- gravitational force
- electroweak force
- strong nuclear force

Newton's Law of Universal Gravitation

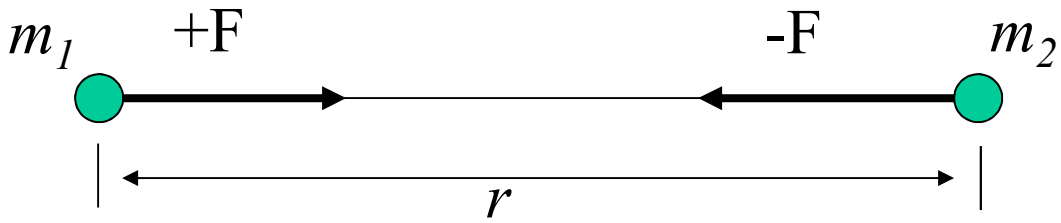
- Every particle in the universe exerts an attractive force on every other particle. For two particles that have masses m_1 and m_2 and that are separated by a distance r, the force that each particle exerts on the other is directed along the line joining the particles and has magnitude given by

where G is a constant.

$$F = G\frac{m_1 m_2}{r^2}$$

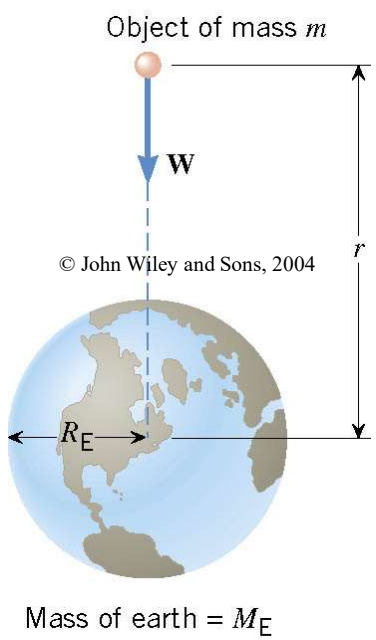
$G = 6.67 \times 10^{-11} Nm^2/Kg^2$

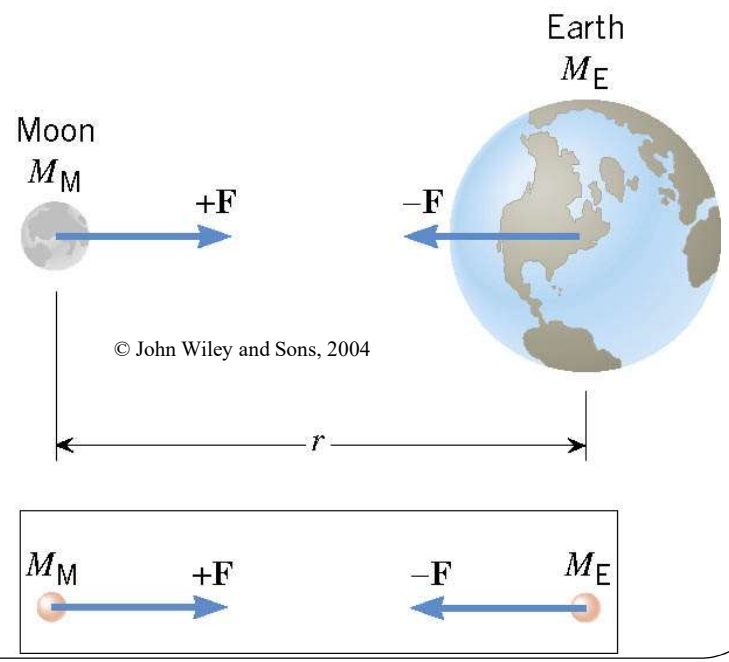
Newton's Law of Universal Gravitation



$$F = G\frac{m_1 m_2}{r^2}$$

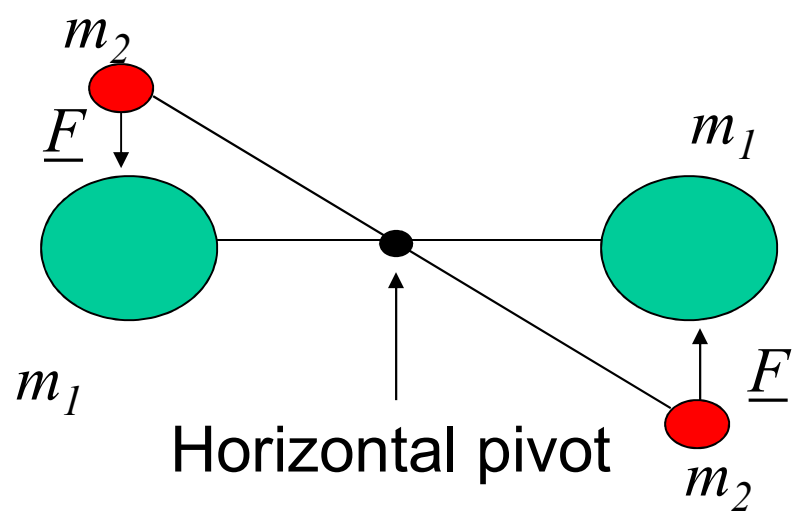
Example of: "inverse-square law"

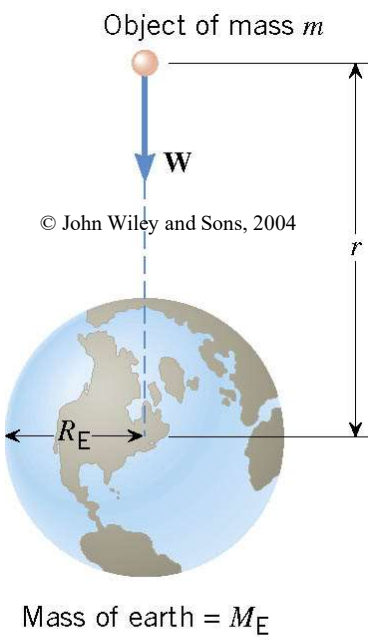




- The value of the constant G was measured by Henry Cavendish in 1798.
  - Measured acceleration of lead balls of known mass due to their mutual attraction.
  - Best value today: (web ref)
- $G = 6.673 \pm 0.019 \times 10^{-11}\ N\ m^2/kg^2$

Torsion balance with lead spheres





Definition of Weight

The weight (W) of an object close to the earth is the gravitational force that the earth exerts on the object.

$$W = G\frac{M_E m}{radius^2}$$

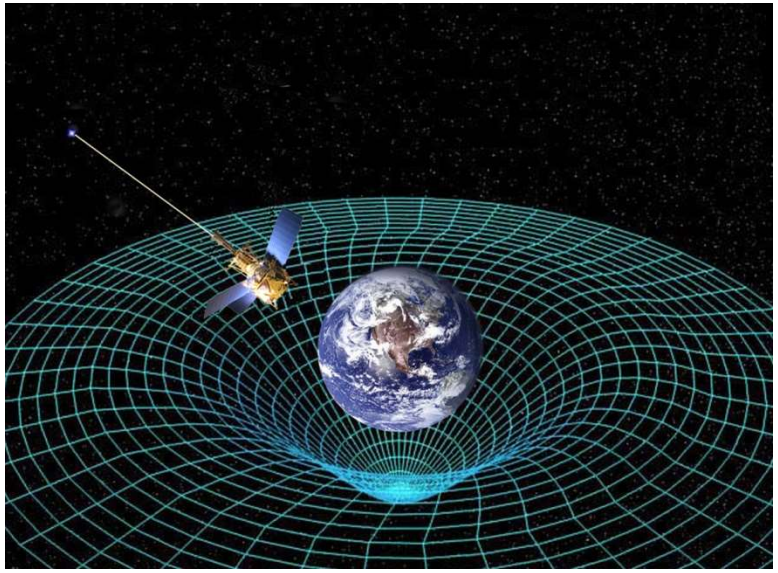
SI unit of weight: newton (N)

By Newton's second law, this force will cause the object to accelerate.

$$W = G\frac{M_E m}{r^2} \quad \text{and} \quad W = ma = mg$$

$$\therefore\ g = G\frac{M_E}{r^2}$$

General relativity - gravity



Summary

- Law of Universal gravitation
- Weight and mass are <u>not</u> the same thing.

$$F = G\frac{m_1 m_2}{r^2}$$

•Weight is force that acts on a body in a gravitational field. Normal force stops us falling through the floor – gives us <u>sensation</u> of weight.